Natural Experiments Show Fuel Subsidies Amplify Fishing Pressure in Industrial Fisheries

Juan Carlos Villaseñor-Derbez[1,2], Christopher Costello[3,4], and Olivier Deschênes[4]

[1]Department of Environmental Science and Policy, Rosenstiel School of Marine, Atmospheric, and Earth Sciences, University of Miami

[2]Frost Institute for Data Science & Computing, University of Miami

[3]Bren School of Environmental Science & Management, University of California Santa Barbara

[4]Department of Economics, University of California Santa Barbara

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Abstract

Fuel subsidies to the fishing industry are widely considered one of the leading drivers of overfishing, prompting global calls for reform. However, rigorous empirical evidence quantifying the behavioral and production responses to subsidy removal remains limited. We address this question by analyzing 14 years of high-resolution data on fisher-level fuel subsidy allocations, fishing activity, and production in Mexico's shrimp trawl fleet. Leveraging year-to-year variation in Mexico's subsidy policy, we find that receiving a fuel subsidy increases fishing effort by 40.6%, with similar responses for fished area and landed catch. Mexico's abrupt subsidy elimination in 2020 led to a 30% reduction in fleet size and decreased fishing time among remaining vessels. Together, these results show that fuel subsidies meaningfully increase fishing pressure and support ongoing global efforts to eliminate fuel subsidies.

Significance: International agreements—including SDG Target 14.6 and Target 18 of the Kunming–Montreal Global Biodiversity Framework—call for major reforms to fisheries subsidies. After decades of negotiation, the WTO has reached a global deal to curb harmful subsidies, yet the behavioral consequences of such reforms remain uncertain. Our study fills this gap by providing plausibly causal estimates of how fishers respond to fuel subsidies. The results inform debates about the likely effects of global reform and, together with related work, suggest that eliminating capacity-enhancing subsidies would substantially reduce fishing effort—both by inducing the exit of marginal vessels and by lowering effort among those that remain active..

1 Introduction

Fisheries subsidies are widely regarded as a major driver of overexploitation. [1]. Scientists, practitioners, and politicians worldwide have called for eliminating or reducing fisheries subsidies as part of a global concerted effort to rebuild fish stocks [2]. However, our ability to predict the social and environmental consequences of such reforms depends on answering two central unresolved questions: How much additional fishing effort do fuel subsidies induce, and where does that effort materialize—across which regions, fisheries, and type of activity? If the amount of overfishing induced by fuel subsidies is large, then the reforms could have large upsides. However, if the amount is small relative to other sources of overfishing (*e.g.* by-catch or illegal, unreported, and unregulated fishing), then it may be better to focus management efforts on addressing those. Similarly, if most "subsidized" effort is spatially located in specific regions or fishing grounds, then we would expect that only those fishing grounds would benefit from a reform [3]. Using high-resolution vessel tracking data from Mexican shrimp trawlers and long-term administrative records on fisher-level subsidy allocations, we quantify how fuel subsidies shape fishing intensity, spatial deployment of effort, and production in Mexico's largest and most valuable fleet.

In standard economic theory, subsidizing an input such as fuel lowers its effective price and leads to inefficiently high input use relative to the social optimum. When the input usage creates an externality (like carbon emissions or overfishing [4, 1]), the subsidy leads to two sources of lost economic efficiency. The first is the usual cost associated with a market distortion, which arises because resources are being misallocated. The second is associated with greater production of the externality itself. Despite limited empirical attention, this issue is pivotal to the sustainability of agriculture, fisheries, mining, and other natural resource sectors and underlies current policy initiatives aimed at reforming or reducing subsidies in these industries [2]. This paper focuses on this second type of deadweight loss in the context of fuel subsidies in industrial fisheries.

Fisheries subsidies are prevalent in most coastal nations, where governments typically spend enormous amounts on subsidy programs [1]. In 2018, global fisheries subsidies totaled approximately USD $35.4 billion, including USD $7.7 billion allocated specifically to fuel support. These large figures have prompted calls for global subsidy reforms [2, 5], and a particular focus has been placed on ending cost-reducing and capacity-enhancing subsidies such as fuel subsidies. Although there is broad consensus about the potential threats and damages posed by fuel subsidies in fisheries, empirical evidence on the magnitude of their social and environmental costs remains limited.

There are two empirical challenges to studying the effect of fuel subsidies on fishing outcomes. The first is data availability: One must observe subsidy allocations, fishing behavior, and/or fisheries production through time, none of which are readily available. And even when they are, a second challenge arises. Governments typically implement blanket subsidy policies for all fishers in a region, which eliminates the quasi-experimental variation required to establish causality in observational studies [6].

However, recent improvements in data and research design have begun to yield credible evidence on this question. For example, Wang et al. [7] studied the interaction between fuel subsidies and vessel buyback programs in China. By observing vessel-level subsidy allocations, they found that reductions in fuel subsidies accelerated the rate at which fishermen retired their vessels. However, they only studied the extensive margin effects of subsidies as they did not employ vessel-tracking data to measure intensive margin effects. Conversely, recent work by Englander et al. [6] leveraged vessel-tracking data to show that fuel subsidies to China's distant water fishing fleet have a large impact on the fleet's fishing effort, and that biological overfishing could be greatly reduced in several high seas regions if China were to reduce fuel subsidies. While they were able to study the intensive and extensive margin effects of fuel subsidies, they did not directly observe vessel-level subsidy allocations. Neither work studied the effects on fisheries production.

To the best of our knowledge, these studies by Wang et al. [7] and Englander et al. [6] are the only ones to provide rigorous causal estimates of the effect of fuel subsidies on fisheries. Important as their contributions are, they focus on China, a nation whose fishing dynamics, magnitude, and scale are so different that most work on fisheries treats them differently [8, 9]. This means their findings may not generalize to other fleets–particularly those in developing nations–highlighting the need to study the effect of fuel subsidies on fisheries in other regions. We contribute to this literature by focusing on Mexico, the world's

13^{th} largest fishing nation[9]. As we will explain in detail below, Mexico provides an ideal empirical setting in which to study the effect of fuel subsidies on fishing behavior and fisheries production.

The paper proceeds as follows. section 2 provides key details on Mexico's fuel subsidy program and explains our identification strategy. section 3 presents our results, which are discussed in section 4. A detailed account of our data and methods are provided in section 5.

2 Mexico's Fuel Subsidy Program

Mexico's well-developed fishing industry has a long history of being subsidized by federal programs, particularly through fuel subsidies [10, 11, 12, 13]. This section provides a brief overview of the main characteristics of Mexico's fuel subsidy program.

Our study period covers 2011 - 2024. During this time, Mexico's fishery management agency (CONAPESCA) maintained a limited-entry roster of economic units (i.e., fishers or fishing companies) eligible to receive a fuel subsidy each year. An economic unit may only "enter" the roster if another unit "exits" it.[1]. Once in the roster, the amount of subsidy granted to an economic unit is a function of their vessel's engine power and the prescribed duration of the fishing season. However, fishery managers can adjust the final subsidy amount each unit receives based on a factor constant across all units that depends on the annual national allocations to the program (For more details, see Supplementary Materials). Subsidized economic units receive money via a government-issued debit card, which can only be used at fueling depots. Any unspent money at the end of the year is reclaimed by CONAPESCA.

The fuel subsidy program was suddenly eliminated in 2020. In late 2019, Mexico's former president signed into law the "Republican Austerity Law", which aimed to curtail public spending. Almost at the same time, in early 2020, the World was faced with the COVID-19

[1]For example, an economic unit could be removed from the roster if they failed to carry a functioning vessel monitoring system or if they failed to report their landings.

pandemic. Together, these two factors forced Mexico to completely eliminate fuel subsidies without notice [13]. Although this led to multiple protests from fishers and increased political pressure[2], the federal government maintained its position, and Mexican fishers have not received subsidized fuel since.

We seek to understand how fuel subsidies modify the *magnitude* and *footprint* of fishing, as these factors determine the potential benefits of subsidy reform. As an example to motivate our analysis, Figure 1 shows how fishing activity by one economic unit changes when they receive an annual fuel subsidy of MXN $231,543 (about USD $12,388). When subsidized, the economic unit increases fishing hours (from 513 hrs/yr to 2,880 hrs/yr) and the extent of its fishing grounds (from 8,395 Km^2 to 12,572 Km^2). While this example from a single economic unit cannot control for confounding factors, it suggests how fuel subsidies influence fishing intensity and spatial distribution.

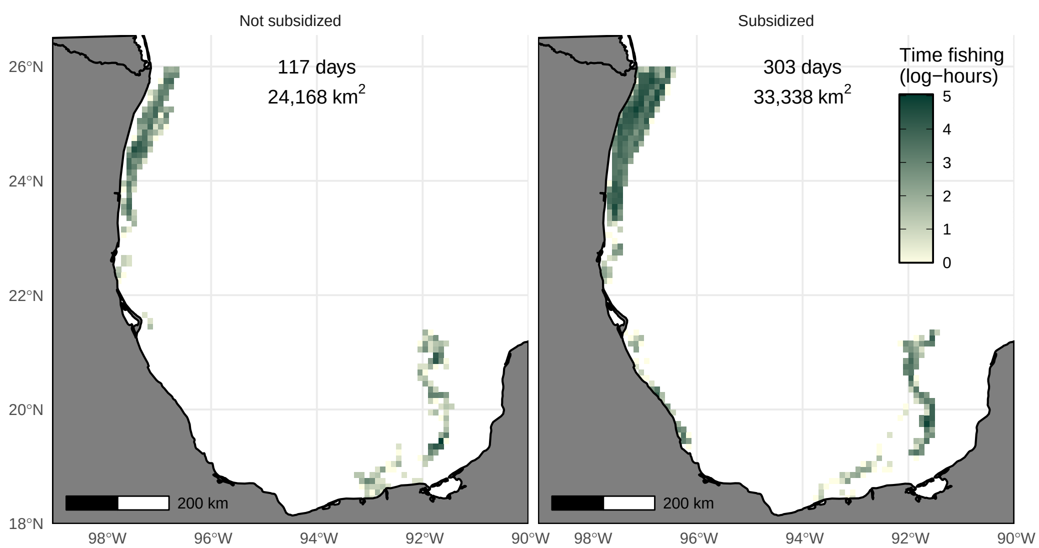

Figure 1: Example of changes fishing behavior in relation to subsidy status. Maps show fishing activity by the same economic unit in a year without a subsidy (left) and a year with a subsidy (right). This fisher spent nearly three times more time fishing when subsidized than when not, and the extent of its fishing grounds is around 50% larger when subsidized than when not. The footprint of fishing effort is shown along a 0.1° grid.

[2]See El Sudcaliforniano: Pega a pescadores la falta de apoyos for a news report and a letter by Senator Cecilia Sánchez García denouncing the removal of fisheries subsidies.

2.1 Identification Strategy

Our empirical strategy leverages three complementary sources of quasi-experimental variation in Mexico's fuel subsidy program. Together, these provide plausibly exogenous shocks to fishing costs that allow us to estimate how fuel subsidies shape fishing behavior and fisheries production. We being by summarizing the core identifying assumptions underlying each source of variation and the corresponding estimation strategy.

Roster turnover and changes in subsidy status. First, we exploit changes in subsidy status generated by the institutional design of CONAPESCA's subsidy roster. Entry into the roster can only occur when another economic unit exits. Because an economic unit cannot directly influence whether others exit the roster, these transitions effectively induce externally driven shifts between *subsidized* and *not subsidized* status for units that are subsidized only in some years. We use this roster turnover to identify how receiving a subsidy affects the probability of fishing, the probability of reporting landings, and the magnitude of fishing activity conditional on participation in the fishery. These effects are estimated using two-way fixed effects regressions that capture semi-elasticities with respect to subsidy status.

Within-unit variation in subsidy amounts. Second, we leverage within-unit variation in subsidy amounts generated by the program's annual "adjustment factor" revisions. This factor is determined by federal budget allocations and administrative decisions that vary across regions and years but are orthogonal to individual fishers' short-run incentives or expected fishing behavior. As a result, the year-to-year fluctuations in subsidy amounts for the same economic unit provide treatment-intensity variation that is plausibly exogenous. For example, Figure S2 shows how economic-unit-level subsidy allocations changed between 2011 and 2019, when Mexico provided fuel subsidies to its fleet. We exploit this within-unit, across-year variation using log–log panel data regressions with economic-unit and region-year fixed effects, yielding elasticities of fishing time, fished area, and landed catch with respect to the amount of subsidy received.

Abrupt elimination of subsidies in 2020. Third, we exploit the sudden and unanticipated elimination of fuel subsidies in 2020 as a natural experiment. The combination of the 2019 Republican Austerity Law and fiscal pressures associated with the COVID-19 pandemic resulted in the complete removal of subsidies without advance notice. Because the reform was nationwide, non-discretionary, and unrelated to the performance or behavior of individual economic units, it provides a clean policy shock. We use this setting to estimate (i) the probability that an economic unit exits the fishery after subsidies disappear and (ii) changes in fishing behavior among units that remain active. These effects are estimated with event-study models and two-way fixed effects regressions.

Together, these three sources of variation allow us to estimate: (i) The percent change in fishing activity with respect to changes in subsidy status; (ii) The percent change in fishing activity associated with a 1% change in the subsidy amount; and (iii) long-run and aggregate responses using the 2020 reform. This multi-layered identification strategy provides a comprehensive assessment of how fuel subsidies affect fishing behavior, spatial footprint, and fisheries production.

Mexico provides a data-rich environment conducive to this study. Using different sources of administrative data, we construct an annual panel data set on subsidy allocations (MXN \$), time fishing (hours), area fished (km^2), and landed catch (kg) by economic unit. This panel allows us to study changes in fishing behavior and production of 920 shrimp trawling vessels owned by 341 economic units. We focus on Mexico's shrimp trawl fleet because they are Mexico's largest and most important fleet, accounting for more than 45% of the value of all Mexican fisheries and catching around 50,000 tons each year [14, 15, 16]. Fuel represents around 4-% of the fleet's variable costs, and subsidies are an important source of profit [17, 13]. The fleet also received between 48% and 67% of the annual fuel subsidies provided to all industrial economic units fishing in Mexico.

A visual inspection of our panel data set reveals three general patterns (Figure 2). First, economic units spend more time fishing, fish a greater area, and land more shrimp than vessels that are not subsidized. Second, economic units that were always subsidized between 2011 and 2019 (N = 142) consistently fish more than those that were only sometimes subsidized (N = 167), and *vice versa*. And third, that the pattern persists even for the subset of economic units whose subsidy status changes in time within our sample (labeled "sometimes").

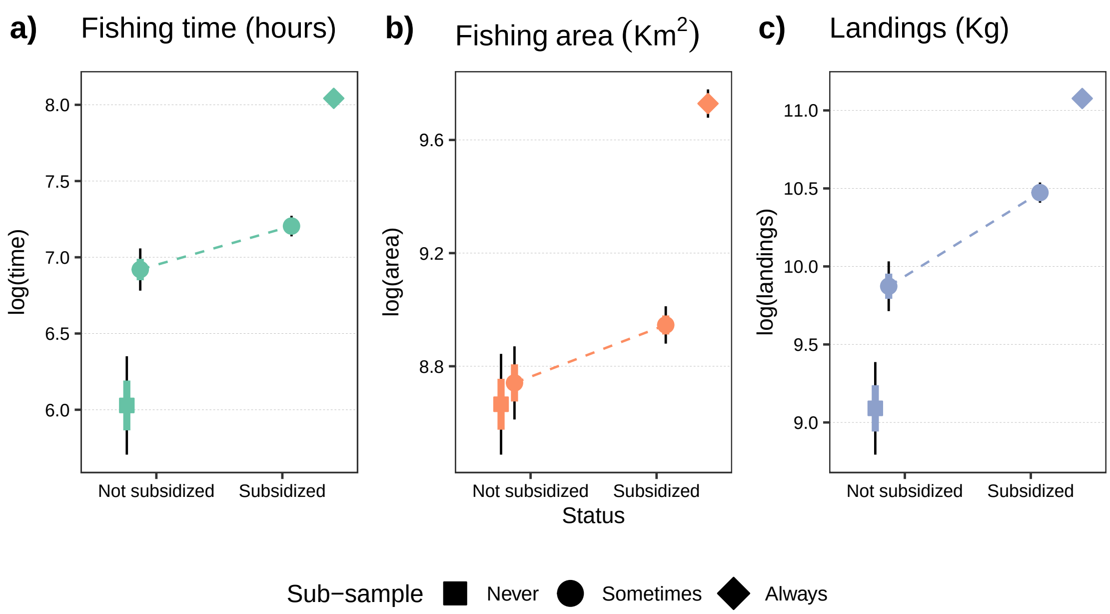

Figure 2: Fishing behavior and fisheries production in relation to subsidy status (2011-2019). The horizontal axis shows the subsidy status and the vertical axis shows the log-transformed outcome of interest (time fishing, area fished, and landings). Points show mean values, error bars show standard errors (colored portion) and 95% confidence intervals (thin black lines). Marker shapes indicate subsidy category with respect to number of times subsidized (never, sometimes, always). The dashed lines connect the mean values for economic units that are sometimes subsidized across subsidized status.

3 Results

We structure our results around the three sources of variation described in subsection 2.1. First, we examine how changes in subsidy status affect fishing behavior and production. We then quantify how these outcomes respond to the amount of subsidy received. Next, we study the effects of Mexico's 2020 reform. Finally, we combine these estimates to measure the aggregate and spatial impacts of fuel subsidies.

3.1 Responses to change in subsidy status

Our sample contains 341 economic units targeting shrimp between 2011 and 2024. Of these, 32 units never received a subsidy, 142 were always subsidized, and 167 were subsidized only in some years. We exploit within–unit changes in subsidy status for this last group to estimate how receiving a subsidy affects participation, fishing activity, and landings.

We begin by examining the extensive margin. Using a two-way fixed-effects regression (see Methods), we find that receiving a subsidy increases the probability that an economic unit engages in fishing by approximately 20%, as measured by both fishing activity and area fished. The probability of reporting landings increases by 41% ($p < 0.01$; Table 1A). These results indicate that some economic units participate in the fishery—and report their catch—only when subsidized.

We then examine whether an economic unit fishes more when a vessel is subsidized. On average, subsidized units fish 350 additional hours per year ($p < 0.01$), expand their fishing grounds by 2,528 km^2 ($p < 0.01$), and land 3.2 additional tons of shrimp ($p < 0.01$). Relative to the mean outcomes of unsubsidized units, these correspond to increases of 24.8%, 36.5%, and 170% in time fishing, fishing area, and landings, respectively (Table 1B).

Finally, we estimate semi-elasticities for economic units that fish regardless of their subsidy status. Receiving a subsidy is associated with a 40.69% increase in time fishing ($p < 0.01$), a 23.5% expansion in fishing grounds ($p < 0.01$), and a 70.18% increase in landings ($p < 0.01$; Table 1C). All results are robust to alternative specifications and sample definitions (see Table S2–Table S4 and Figure S5–Figure S10).

Table 1: Effect of receiving a fuel subsidy on fishing behavior and fisheries production.

	Fishing time	Fishing area	Landings
A) Extensive margin			
Subsidized	0.223 (0.028)***	0.219 (0.029)***	0.413 (0.037)***
N_{eu}	167	167	167
N	1431	1431	1431
R^2 Adj	0.531	0.547	0.597
B) Intensive margin (levels)			
Subsidized	350.248 (65.644)***	2528.599 (523.001)***	32347.423 (6977.661)***
$\bar{Y}_{\text{Subsidized}=0}$	1411	6939	18968
N_{eu}	167	167	167
N	1431	1431	1431
R^2 Adj	0.924	0.792	0.710
C) Semi-elasticities			
Subsidized	0.341 (0.073)***	0.211 (0.068)***	0.532 (0.080)***
N_{eu}	134	127	117
%Change	40.69%	23.50%	70.18%
N	1290	1235	1192
R^2 Adj	0.726	0.700	0.757

$*p < 0.1, **p < 0.05, ***p < 0.01$

The unit of observation is an economic unit by year. Numbers in parentheses are cluster-robust standard errors, clustered at the economic-unit level. Panel A) shows estimates for the extensive margin, where the outcome variables indicate whether a vessel spent time fishing, had fishing grounds, or reported landings. Panel B) shows estimates for the intensive margin, where the outcome variables are time fishing (hours), fishing area (km^2), and landings (kg). Panel C) shows semi-elasticity estimates for log-transformed time fishing (hours), fishing area (km^2), and landings (kg). This last panel excludes observations with fishing activity or landings at exactly zero, mostly capturing the intensive margin.

3.2 Responses to change in subsidy amount

The effects above capture differences between subsidized and unsubsidized economic units. We now examine how fishing effort, spatial footprint, and landings respond to differences in the *amount* of subsidy received.

A key feature of Mexico's fuel subsidy program is that the annual subsidy amount allocated to each economic unit varies across years (see Figure S2). This variation arises from changes in federal budget allocations and administrative decisions within CONAPESCA, including allocations to other programs such as aquaculture subsidies [11, 13]. Because these adjustments are driven by budgetary conditions rather than unit-specific factors, they generate plausibly exogenous within–economic-unit variation in subsidy amounts. We use this year-to-year variation to estimate intensive-margin responses for the 297 economic units that were subsidized at least twice between 2011 and 2019.

Using a two-way fixed-effects regression, we estimate elasticities of fishing time, fished area, and landings with respect to subsidy amount. We find that a 1% increase in the subsidy received is associated with a 0.14% increase in time fishing ($p < 0.01$), a 0.15% increase in fished area ($p < 0.01$), and a 0.20% increase in landings ($p < 0.01$). These estimates are robust to alternative sample definitions. Specifications that substitute covariates for fixed effects produce 40–80% larger elasticity estimates, but the qualitative patterns remain unchanged (see Table S5, Figure S11, and Figure S12).

Table 2: Elasticity estimates for time fishing (hours), fishing area (km^2), and landings (kg) with respect to changes in subsidy amount.

	Fishing time	Fishing area	Landings
log(subsidy amount[MXP])	0.139 (0.021)***	0.147 (0.022)***	0.198 (0.018)***
%Change	0.14%	0.15%	0.20%
N_{eu}	297	292	295
N	2240	2202	2246
R^2 Adj	0.850	0.818	0.876

$*p < 0.1, **p < 0.05, ***p < 0.01$

The unit of observation is an economic unit by year. All models include fixed effects by economic unit and by region-year. Numbers in parentheses are cluster-robust standard errors, clustered at the economic-unit level. The sample contains economic units subsidized at least twice and with subsidy amount > 0. The number of economic units used in each column is shown by N_{eu}.

3.3 Responses to an *impromptu* reform

While the previous subsections rely on within-program variation, Mexico's abrupt 2020 reform provides an opportunity to study how economic units respond when subsidies disappear entirely for a longer period of time. The COVID-19 pandemic interacted with an ongoing austerity plan, resulting in sweeping fiscal and administrative adjustments [12, 13]. The fuel subsidy program was eliminated practically overnight, prompting protests from fishers who argued they could not continue operating without support [18].

We begin by examining exit behavior. We focus on the 142 economic units that were always subsidized prior to the reform. All were active between 2011 and 2019, yet 53 failed to participate in the fishery at least once between 2020 and 2024, and 42 did not return at all by 2024, suggesting permanent exit. Estimating the probability of exit after 2020, we find that the average annual probability of an economic unit leaving the fishery increased by 7.5 percentage points in the post-reform period ($p < 0.01$; Table 3). By 2024, the cumulative exit probability reached approximately 30% (Figure 3).

We next examine how remaining economic units adjusted their behavior. Among units that continued fishing after the reform, we estimate modest reductions along the extensive margin (Panel A of Table 4). Intensive-margin responses were substantially larger: average

Table 3: Effect of Mexico's *impromptu* fuel subsidy reform on probability of economic units exiting the fishery.

	(1)
Post	0.075 (0.011)***
N	1914
R^2 Adj	0.049

$*p < 0.1, **p < 0.05, ***p < 0.01$

The unit of observation is an economic unit by year. Numbers in parentheses are cluster-robust standard errors, clustered at the economic-unit level.

annual fishing time fell by 4,200 hours, fished area declined by 13,900 km^2, and landings decreased by 45.6 tons (Panel B of Table 4). Event-study estimates (Figure 3) show that these changes stabilized around 2022.

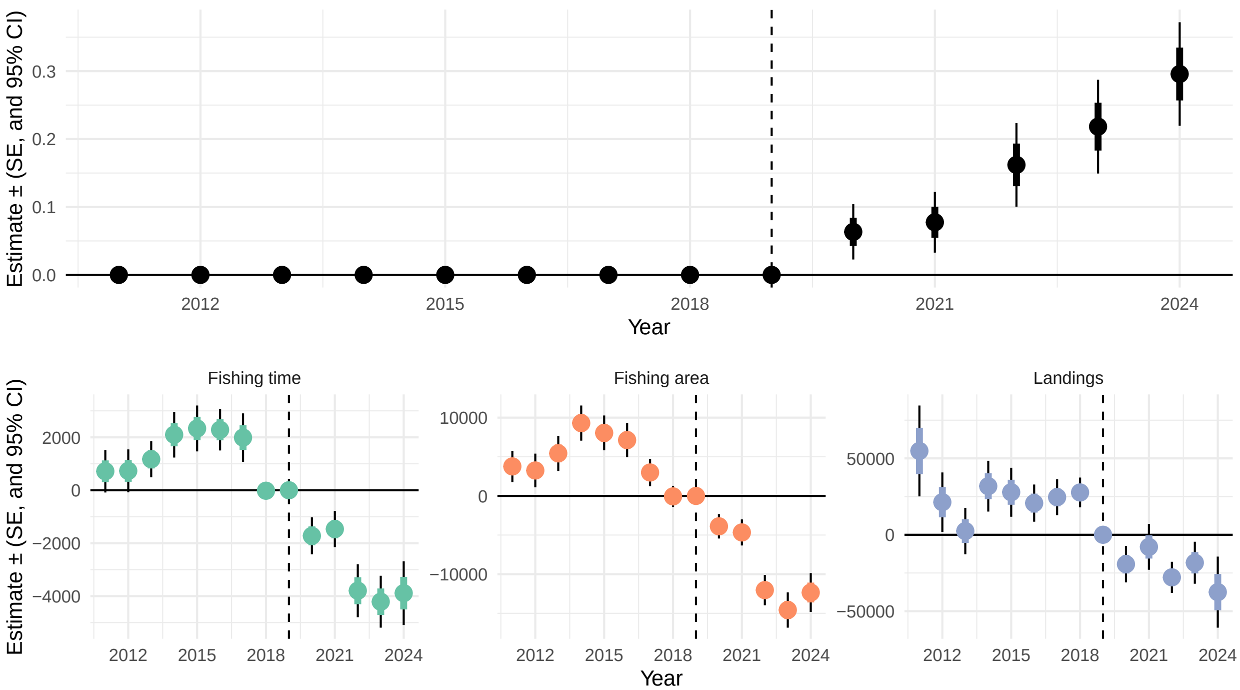

Figure 3: Annual estimates of the effect of Mexico's fuel subsidy reform on fisheries. The top panel shows changes in the probability of exiting the fishery for economic units that were always subsidized between 2011 and 2024. The bottom panels show changes in total fishing time, fishing area, and landings for economic units that did not exit the fishery as of 2024. Points are coefficient estimates, thick colored lines show standard errors, and thin black lines show 95% confidence intervals.

Table 4: Effect of Mexico's *impromptu* fuel subsidy reform on fishing behavior and fisheries production for economic units that did not exit the fishery.

	Fishing time	Fishing area	Landings
A) Extensive Margin			
Post	-0.036 (0.012)***	-0.093 (0.021)***	-0.086 (0.021)***
N	1400	1400	1400
R^2 Adj	0.099	0.185	0.188
B) Intensive Margin (levels)			
Post	-4273.615 (685.285)***	-13931.084 (1112.424)***	-45651.016 (10855.586)***
$\bar{Y}_{\text{Post}=0}$	4472	18351	107863
N	1400	1400	1400
R^2 Adj	0.783	0.748	0.868

$*p < 0.1, **p < 0.05, ***p < 0.01$

The unit of observation is an economic unit by year. Numbers in parentheses are cluster-robust standard errors, clustered at the economic-unit level. Panel A) shows estimates for the extensive margin, where the outcome variables indicate whether a vessel spent time fishing, had fishing grounds, or reported landings. Panel B) shows estimates for the intensive margin, where the outcome variables are time fishing (hours), fishing area (km^2), and landings (kg).

3.4 Aggregate effects of fuel subsidies

We have shown that subsidized economic units fish more, and that the amount of additional fishing increases with the amount of subsidy received. We now examine what these *individual* responses imply for *aggregate*, fishery-wide impacts. In particular: How much of total historical effort is attributable to fuel subsidies? Because vessels are heterogeneous in behavior and spatial footprint, and subsidies are not evenly distributed, we combine yearly vessel-level data on subsidy receipt, subsidy amount, and fishing location to quantify the share of historical fishing activity and landings attributable to subsidies. We then identify areas disproportionately subject to subsidized effort.

3.4.1 Historical impacts of subsidies

Total annual fishing activity (or landings) can be divided into three categories: (1) activity by units that were never subsidized; (2) activity by subsidized units that would have occurred even without the subsidy; and (3) activity by subsidized units that is attributable to the subsidy. Between 2011 and 2019, Mexican shrimp trawlers fished an average of 1.13 $\pm$ 0.15 million hours, of which 1.05 $\pm$ 0.15 million were by subsidized units (Figure 4a). Using our semi-elasticity estimates, we find that 0.42 $\pm$ 0.06 million hours per year can be attributed to fuel subsidies, implying that subsidies accounted for 37.6% $\pm$ 2.4% of total annual fishing hours.

The fleet landed an average of 17.6 $\pm$ 1.6 thousand tons of shrimp per year, with 16.9 $\pm$ 1.8 thousand tons landed by subsidized units (Figure 4c). Our estimates indicate that 11.9 $\pm$ 1.2 thousand tons of landings—about 67% of all shrimp landed by the studied fleet—were attributable to subsidies.

We repeat this exercise using our elasticity estimates to calculate the *percent* reduction in fishing time and landings under alternative subsidy-reduction scenarios (10, 30, 50, and 90%). For instance, capping subsidy allocations at 50% of historical levels would have reduced fishing time by approximately 96.3 thousand hours per year and landings by 2.18 thousand tons (Figure 4b,d).

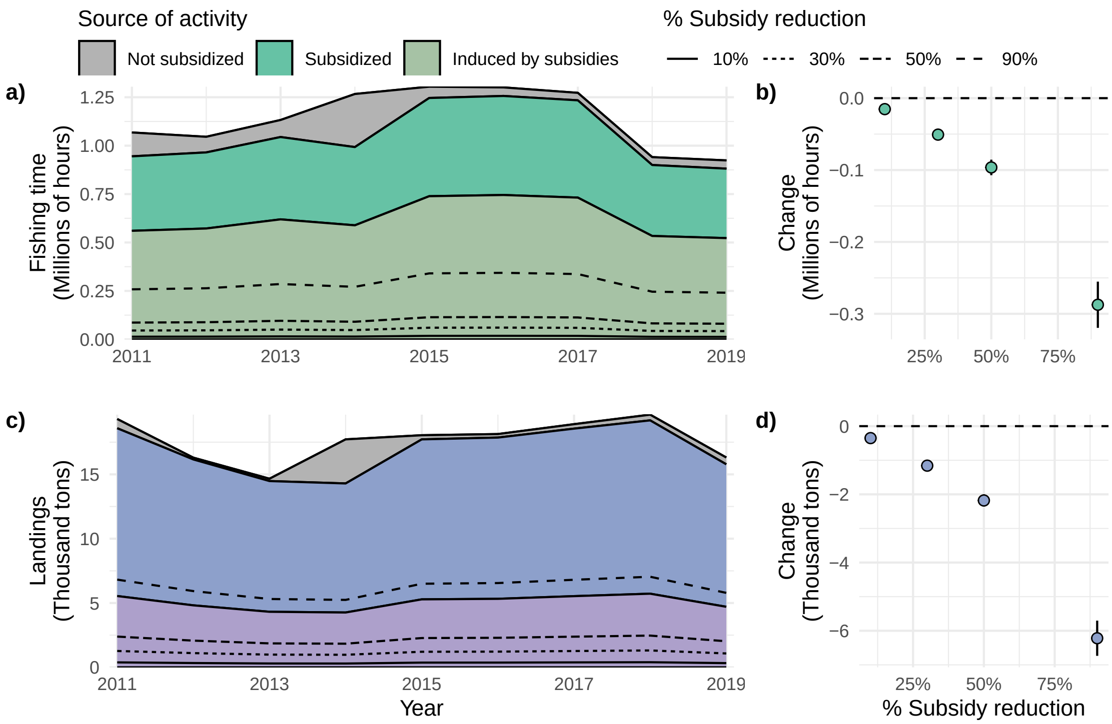

Figure 4: Aggregate effects of fuel subsidies on fishing activity, fished area, and landings. Panels **a)** and **c)** show area-stacked time-series of fishing time and landings[3]. The gray portion is activity and production from economic units that were not subsidized in a given year. The colored portion corresponds to activity and production by economic units that were subsidized. The bottom stack of each panel shows the portion of effort or production by subsidized economic units that is attributable to the subsidy, as indicated by our semi-elasticity estimates (Table 1C). The different line types show the portion of effort that could have been removed had the subsidies been reduced by different amounts. Panels **b)** and **d)** show the mean annual reduction in fishing time and landings expected from four different subsidy reduction policies (estimated as mean of all activity between 2011-2019). Black error bars show 95% confidence intervals and the colored portion shows standard errors.

3.4.2 Spatial implications of a non-spatial policy

The above thought experiments cannot be conducted for fished area because the metric is not additive across economic units. However, understanding the spatial implications of a non-spatial policy is crucial, as the distribution of fishing effort can strongly influence environmental impacts [19]. For example, if most economic units fish in a small subset of grounds, eliminating fuel subsidies would generate large, localized upsides. Conversely, if subsidized and unsubsidized vessels operate in broadly similar areas, subsidy reform would yield more modest but spatially widespread effects. This tension between large and local *versus* modest and widespread benefits raises a natural question: is subsidy-induced fishing effort homogeneously distributed in space?

To answer this, we use our semi-elasticity estimates to simulate counterfactual fishing activity in the absence of subsidies along a $0.1^\circ \times 0.1^\circ$ grid (roughly 11 km by 11 km at the equator). Pixels fished only by unsubsidized units show no change, while those fished exclusively by subsidized units exhibit the largest reductions. Using data from the final year of subsidies (2019, with 286 active units, 247 of which were subsidized), we find that subsidized fishing activity is heterogeneously distributed in space, but that this pattern broadly matches the baseline distribution of activity by unsubsidized units (Figure 5a–b).

Mexico divides its waters into six broad management regions. The Gulf of California (region II) and Gulf of Mexico (region V) sustain the highest levels of both total and subsidized activity (Figure 5a–b). Subsidized effort accounted for up to 40% of total activity along the eastern margin of the Gulf of California and the Campeche Bank, between regions V and VI (Figure 5c). These areas also consistently ranked among the top 90% of pixels where subsidized effort represented a large share of total fishing (Figure 5d).

Finally, the analysis reveals that several areas of particular conservation concern were fished primarily by unsubsidized units. For example, all activity within and around the recently protected Alacranes Reef [20] (region VI) was attributable to unsubsidized vessels. Similarly, fishing in the upper Gulf of California—region II, habitat of the critically endangered Vaquita [21, 22]—was also predominantly conducted by unsubsidized units.

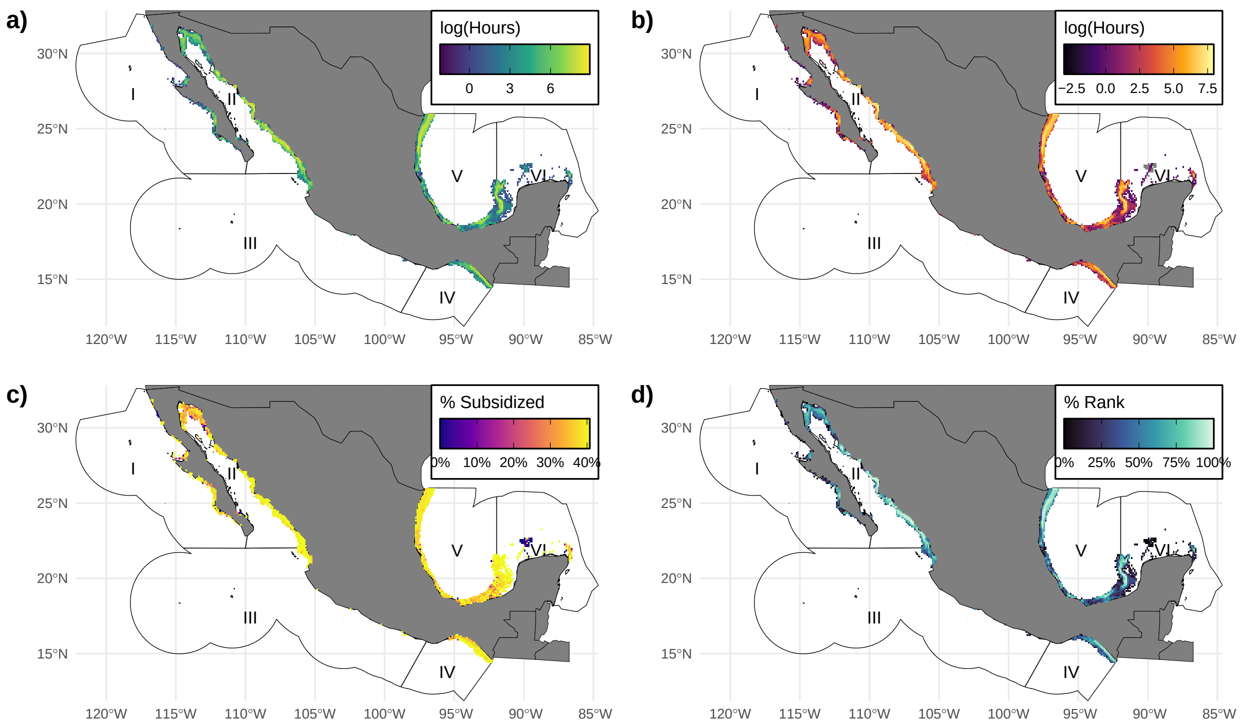

Figure 5: Spatial distribution of the effects of fuel subsidies on fishing activity. Panel a) shows a map of total fishing hours for 2019. Panel b) shows a map of total fishing hours attributable to fuel subsidies. Panel c) shows the per cent of fishing effort that is attributable to fuel subsidies, and panel d) shows the percentile ranking of each pixel. Polygons in the ocean show Mexico's Exclusive Economic Zones, divided into six management regions utilized by Mexico's fishery management agency.

4 Discussion

Our results show that fuel subsidies substantially increase fishing effort, spatial footprint, and landed catch in Mexico's industrial shrimp fleet: First, receiving a subsidy induces large increases in time spent fishing, area fished, and landings. Second, the amount of subsidy matters: fishing effort, fishing grounds, and landings all scale positively with the size of the subsidy allocation. Third, Mexico's abrupt 2020 reform led to substantial exit from the fishery and to marked reductions in effort and landings among vessels that remained active. Taken together, these findings indicate that fuel subsidies have meaningful and measurable fishery-wide impacts.

These results are consistent with theoretical predictions that subsidizing variable inputs increases effort and extraction, though empirical evidence on these predictions remains scarce. Our study adds new evidence using detailed vessel-tracking data, fisher-level subsidy allocations, and policy variation in Mexico's fishery management system. The Agreement on Fisheries Subsidies at the World Trade Organization, which entered into force in 2025, aims in part to curtail precisely these capacity-enhancing subsidies. Our results suggest that reforming fuel subsidies can meaningfully reduce overfishing pressure in industrial fleets.

Our study has limitations common to observational research. The estimates reported in this study can be interpreted as causal effects under two conditions: first, that annual subsidy variation reflects federal budget processes rather than unobserved fisher characteristics, and second, that VMS-measured effort is largely immune to manipulation. These assumptions are more plausible for effort and spatial extent than for landings, where reported catch may be influenced by strategic considerations tied to subsidy eligibility.

Our results show that subsidizing fuel alters fishing activity, but how managers allocate and disburse subsidies also shapes the magnitude of fishers' responses. Mexico's program limits the quantity of subsidized fuel any fisher can obtain because, although there is considerable year-to-year and fisher-to-fisher variation, the allocation rule establishes a 2-peso per liter price subsidy over the first 40–70% of anticipated fuel consumption (DOF 2010–2018).

These subsidies are disbursed as lump-sum transfers that can only be used for fuel. Fishers use this cash to pay for fuel until funds are exhausted (*i.e.*, the first consumed liters are effectively "free"). This price structure resembles an increasing block rate pricing scheme, often used for electricity and water. In those markets, consumers respond to the "average price" rather than the marginal price [23]. Using the allocation rule and a median diesel price of 16.2 pesos per liter, we calculate that Mexico's fuel subsidies reduce the average price of fuel by 4.9–8.6%, consistent with other estimates (*e.g.*, 8.2% in Revollo-Fernández et al. 12). Our empirical results suggest that this is sufficient to induce a behavioral response.

Our aggregate calculations show that up to 37% of historical fishing effort in Mexico's shrimp trawl fleet is attributable to subsidies, with some areas (*e.g.*, the Campeche Bank and eastern Gulf of California) disproportionately affected. These findings imply that subsidy reform might generate large but localized environmental benefits. Limited availability of stock assessment data precludes precise statements about all relevant stocks, but we can put these numbers in context. For example, the biomass of the heavily fished blue shrimp (*Litopenaeus stylirostris*) stock in the Gulf of California [24] is estimated at 30% below the biomass that would yield MSY (*i.e.*, $B/B_{MSY} = 0.7$). It is therefore reasonable to expect that reducing fuel subsidies would support stock rebuilding, at least along the eastern Gulf of California.

We also show that areas of particular conservation concern (such as Alacranes Reef and the Upper Gulf of California) are primarily targeted by unsubsidized units. This suggests that subsidy reform alone would have limited direct implications for these areas. Other management measures, such as fully protected marine reserves, may be more appropriate when the objective is to reduce fishing pressure over sensitive habitat.

Overall, our findings suggest that subsidy reforms elsewhere may produce spatially uneven responses, with some areas benefiting more than others. It is also important to consider social implications, as some ports or fishing communities may depend more heavily on subsidies. Previous work in Mexico and elsewhere has shown that even ideal management aimed at maximizing long-term yields may not raise fishers' income above the poverty line [25, 26]. Some have suggested reallocating funds from harmful fuel subsidies toward social programs intended to raise fishers' income [27], although practical pathways remain unclear. This tension between biological gains and political costs may help explain many nations' hesitation to reform fisheries subsidies and highlights the importance of understanding the distributional implications of reform.

We conclude that fuel subsidies induce overfishing, that the magnitude of overfishing is non-trivial, and that its spatial effects are highly localized. These findings support calls for subsidy reform [2] but also underscore the need for realistic expectations. Our results are directly relevant to Mexico and to other coastal nations considering reducing or eliminating fuel subsidies to their industrial fishing fleets.

5 Methods

We are interested in studying how fuel subsidies affect fishing activity and production. Here, our unit of observation are "economic units", a term used by the Mexican fisheries agency (CONAPESCA) to refer to an individual or firm that participates in a fishery. We combine administrative datasets on subsidy allocations by economic unit and vessel tracking data to construct a unique panel of annual subsidies and fishing activity by economic unit. The following subsections provide further details on data procurement, filters, and sample construction.

5.1 Datasets and their sources

We make use of six types of data to study the effects of fuel subsidies on fishing behavior. Historical subsidy allocations and a vessel registry allow us to identify subsidized economic units and their characteristics. Then, we use vessel tracking data and historical fisheries production data to derive our three main outcomes of interest: fishing time, fishing extent, and fisheries production. Finally, we also use historical diesel fuel prices and monthly indices for El Niño Southern Oscillation to include fuel costs and environmental variation as covariates. Each of these is described in detail below.

5.1.1 Subsidy allocations

Data on subsidy allocations to each economic unit come from CausaNatura, an NGO whose mission is to compile, procure, and make available administrative datasets relevant to environmental and natural resource management. We use the "*Padrón de beneficiarios de Combustibles*", which was last updated on June, 2020. This administrative dataset contains information on the annual subsidy cap assigned to economic units fishing in Mexico during the 2011 - 2019 period (n = 4,597). From this information, we assign treatment status (subsidized or not subsidized) to all economic units in our sample (see subsection 5.2), and the amount of fuel subsidy received by each.

5.1.2 Vessel registry

We use an official vessel registry with information for all large-scale fishing vessels that hold a fishing permit in Mexico, which was also provided by CausaNatura. The vessel registry includes unique vessel identifiers and economic unit identifiers (ownership), vessel dimensions (length overall, beam, draught, and gross tonnage), species-specific fishing permits granted, and engine characteristics (*e.g.* total engine power, type of fuel used by the engine, and engine model). The registry contains information for 3,093 vessels owned by 1,093 economic units. From these, 1,415 are licensed to use bottom trawl nets and 1,561 are licensed to participate in the shrimp fishery; 1,368 are licensed to use both (and are owned by 464 economic units).

5.1.3 Vessel tracking data

[28, 29] There are two general types of vessel tracking technologies: Automatic Identification Systems (AIS) and Vessel Monitoring Systems (VMS). AIS is designed as a vessel-to-vessel broadcast system intended to help avoid at-sea collisions between vessels [8]. VMS, on the other hand, is employed by governments to track vessels of interest, and a vessel's position is broadcast directly to a central repository instead of to other vessels [30]. We use VMS data from Mexico's satellite monitoring system of fishing vessels (*i.e.* SISMEP[28]). These data are publicly available and continuously updated at datos.gob.mx. The version we use was downloaded on June 15, 2024. These VMS data contain the timestamp, geographic location (latitude and longitude), and speed of 2,775 vessels between January 1, 2007 and Feb 29, 2024.

Regulations require all fishing vessels larger than 10.5 m in length overall and with an in-board motor of more than 80 horsepower to carry a Vessel Monitoring System (VMS)[4].

[4] Regulatory text available at: https://www.monitoreodeembarcaciones.com.mx/monitoreosatelital/QuienDebe.htm

Failure to comply with this VMS requirement automatically disqualifies a vessel as eligible to receive any type of subsidy.

5.1.4 Fisheries production

[31, 32] Fisheries production data come from Mexico's landing receipts, where fishers report their landings. As with the VMS requirement, failure to report catch makes a fisher ineligible to receive a subsidy. The dataset contains information on the identity of the economic unit and vessel landing the catch, the target species, and the amount (Kg).

5.1.5 Fuel prices

We also compile price data for diesel fuel used by these economic units by combining two sources of information. The first one is reported by the Energy Information System ("Sistema de Información Energética"; "SIE") and contains the national annual average price of diesel between 2011 - 2016, when fuels were subject to nation-wide price controls. Price controls were lifted in 2017, and fuel prices were determined by local supply and demand. The Energy Regulatory Commission ("Comisión Reguladora de Energía"; CRE) reports monthly state-level prices after 2017, which we use to calculate annual national averages for 2017 - 2019 period. We use Mexico's consumer price index reported by the OECD to adjust prices to 2019 Mexican pesos.

5.1.6 Environmental covariates

The productivity of shrimp fisheries is known to be influenced by ENSO events [33]. We use the Mean NINO 3.4 index available from NOAA's Physical Sciences Laboratory Climate Indices repository (Monthly Atmospheric and Ocean Time Series). We use monthly means to produce an annual mean value of ENSO 3.4, which we include as a time-varying covariate in some of our regressions.

5.2 Data processing

5.2.1 Sample construction

We limit our data to activity occurring between 2011 and 2019, the years for which subsidy allocation data are available. Additionally, retain vessel tracks occurring at depths between 0.15 and 100 m deep (as indicated by GMEDs bathymetric dataset) because shrimp trawlers in Mexico are not allowed to fish shallower than 9.15 m deep and they operate at a maximum depth of 100 m [14]. Shrimp trawlers typically operate speeds between 1 and 5 knots[5], so we also filter tracks based on their speed. These filters result in a total of 1,177 vessels belonging to 414 economic units. We further restrict the sample to economic units that are only licensed to fish for shrimp using trawl nets, leaving us with 409 economic units.

5.2.2 Outcomes of interest

Our first outcome of interest is fishing activity. We define it as time (hours) a vessel spent traveling at speeds between 1 and 5 knots in areas between 9.15 and 100 m depth. We calculate an economic unit's total annual fishing hours as the sum of fishing hours across all their vessels.

Our second outcome of interest is the total extent of fishing grounds (km^2) in which these economic units operate. We used a density-based spatial clustering algorithm (DBSCAN) to identify fishing grounds based on individual vessel positions. The algorithm was applied to all positions at the vessel-by-year level. The algorithm clusters points based on their distribution across space, given a minimum number of points per cluster and a maximum distance between points. We used a maximum distance of 50Km and a minimum of 6 points per cluster. Clusters thus represent the group of individual GPS coordinates that are associated with a fishing ground. Points without cluster membership were dropped. We then built a convex hull around each cluster and calculated its area. The total extent of fishing

[5] Catálogo de los Sistemas de Captura de las Principales Pesquerías Comerciales, available at: CONAPESCA

grounds of an economic unit was then calculated as the sum of all fishing grounds used by their vessels. For this portion of the analysis, geographic coordinates were reprojected onto a Mexico Lambert Conic Conformal projection (With EPSG code 6361).

Our third and last outcome of interest is the total amount of catch landed by each economic unit, which we derive from our fisheries production dataset. Our sample is therefore made up of large-scale economic units that target shrimp and carry VMS transponders. This group receives between 48.22% and 67.73.% of the annual subsidies awarded to all industrial economic units fishing in Mexico. The final estimation sample is a panel of annual economic-unit fuel subsidy allocations (in 2019 MXP), time, extent, landings, and control variables such as fuel prices, total horsepower of number of active vessels owned by an economic unit, and environmental indices (*i.e.* NINO3.4 index). These data contain 3,376 observations attributed to 409 economic units between 2011 and 2019. Tables with summary statistics are included in the supplementary materials (Table S1).

5.3 Empirical strategy

5.3.1 Changes in subsidy status

Subsidy allocations are uncorrelated with the outcomes of interest (fishing hours, fishing area, and fisheries production) so we can use these quasi-random changes in subsidy status to test for changes in fishing behavior and fisheries production for economic units whose subsidy status changed at least once in our study period (2011-2019). We estimate the semi-elasticity (*i.e.* the % change in outcome of interest caused by change in subsidy status) of time fishing, fished area, and landings with respect to subsidy status in a two-way fixed-effects regression framework.

$$\mathbb{1}(y_{it} > 0) = \beta S_{it} + \omega_i + \lambda_{r(i)t} + \epsilon_{it} \tag{1}$$

Where $\mathbb{1}(y_{it} > 0) = 1$ if economic unit i has fishing hours, fished area, or landings reported for year t. Then, S_{it} is a dummy variable indicating subsidy status, ω_i and $\lambda_{r(i)t}$ are fixed-effects by economic unit and by region-year. ($r(i)$ denotes the (time-invariant) region to which economic unit i belongs). Our parameter of interest is β, which captures the extensive margin effects. We also estimate model where our outcome variables are simply y_{it} and $log(y_{it})$, in which case β yields changes along the intensive margin and a semi-elasticity, respectively. Our supplementary materials include alternative specifications and sample definitions (See subsection A.3).

5.3.2 Changes in subsidy amount

Recall that our dataset has three types of economic units: those that were never subsidized, those that were subsidized at least one year, and those that were subsidized every year in our dataset. For the later two types, the *amount* of subsidy they receive varies by year (See Figure S2). This annual variation is due to budgetary constraints, which arise when CONAPESCA receives different amounts of funding in the annual federal budget or when funds are allocated to other programs (*e.g.* aquaculture development). We can use these quasi-random changes in subsidy amounts to test for changes in fishing behavior and fisheries production for economic units who were subsidized at least twice between 2011-2019. Like before, we estimate our coefficient of interest (this time an elasticity) in a two-way fixed effects framework with our estimating equation taking the form:

$$log(y_{it}) = \beta \log(s_{it}) + \omega_i + \lambda_{r(i)t} + \epsilon_{it} \tag{2}$$

Where $log(y_{it})$ is the log-transformed outcome of interest and s_{it} is the subsidy amount allocated to economic unit i in year t, in 2019 Mexican pesos. ω_i and $\lambda_{r(i)t}$ are fixed-effects by economic unit and by region-year. The parameter of interest is β, which captures the elasticity of fishing with respect to subsidy amount. Our robustness checks for this analysis (See subsection A.3) test for changes in the estimated coefficient when limiting the sample to vessels subsidized at least 3, 4... 8 times (Figure S10) or where we use different specifications

(Figure S11).

5.3.3 An *imromptu* reform

Mexico eliminated fuel subsidies starting in 2020. We first test for changes in the probability of a vessel exiting the fishery using the following specification:

$$E_{it} = \beta \text{Post}_t + \omega_i + \epsilon_{it} \tag{3}$$

Where E_{it} is an indicator variable that takes a value of $E_{it} = 1$ if economic unit i exited the fishery at year t and is $E_{it} = 0$ otherwise. The variable Post_{it} takes a value of $\text{Post}_t = 1$ if the observation comes from the post-reform period (2020 and later) and $\text{Post}_{it} = 0$ otherwise. Finally, ω_i are fixed-effects by economic unit. The parameter of interest is β, which captures the average annual probability of a vessel exiting the fishery in the post-reform period. We are also interested in understanding how the probability of exiting the fishery evolved in the post-reform period. We therefore estimate a model of the form:

$$\mathbb{1}(y_{it} = 0) = \sum_{t \neq 2019} \beta_t \text{Year}_t + \omega_i + \epsilon_{it} \tag{4}$$

Where $\mathbb{1}(y_{it} = 0)$ Indicates whether an economic unit exits the fishery in year t. Year_t are year indicators (omitting 2019 as the reference category). In this case the parameter of interest, β_t, captures the average probability of an economic unit exiting by year t.

We then use the same model specifications, but this time with outcomes $\mathbb{1}(y_{it} = 0)$ and $(y_{it} = 0)$ that measure extensive and intensive margin changes, respectively. This time the sample is restricted to economic units that did not leave the fishery.

References

[1] U R Sumaila, L Teh, R Watson, and others. Fuel price increase, subsidies, overcapacity, and resource sustainability. *ICES J. Mar. Sci.*, 2008.

[2] U R Sumaila, L Alam, P R Abdallah, D Aheto, and others. WTO must complete an ambitious fisheries subsidies agreement. *npj Ocean*, 2024.

[3] Enric Sala, Juan Mayorga, Christopher Costello, David Kroodsma, Maria LD Palomares, Daniel Pauly, U Rashid Sumaila, and Dirk Zeller. The economics of fishing the high seas. *Science advances*, 4(6):eaat2504, 2018.

[4] F L V Machado, V Halmenschlager, P R Abdallah, and others. The relation between fishing subsidies and CO2 emissions in the fisheries sector. *Ecological*, 2021.

[5] World Trade Organization. WTO agreement on fisheries subsidies enters into force.

[6] Gabriel Englander, Jihua Zhang, Juan Carlos Villaseñor-Derbez, Qutu Jiang, Mingzhao Hu, Olivier Deschenes, and Christopher Costello. Input subsidies and the depletion of natural capital: Chinese distant water fishing. *Journal of Environmental Economics and Management*, 130:103127.

[7] Kaiwen Wang, Matthew N Reimer, and James E Wilen. Fisheries subsidies reform in china. 120:e2300688120.

[8] David A Kroodsma, Juan Mayorga, Timothy Hochberg, Nathan A Miller, Kristina Boerder, Francesco Ferretti, Alex Wilson, Bjorn Bergman, Timothy D White, Barbara A Block, and Others. Tracking the global footprint of fisheries. *Science*, 359(6378):904–908, 2018.

[9] FAO. *The State of World Fisheries and Aquaculture 2024*. FAO ;, 2024.

[10] Andrés M Cisneros-Montemayor, Enrique Sanjurjo, Gordon R Munro, Victor Hernández-Trejo, and U Rashid Sumaila. Strategies and rationale for fishery subsidy reform. *Mar. Policy*, 69:229–236, July 2016.

[11] Virginia Leal Cota and José Eduardo Rolón Sánchez. Análisis del ejercicio de los subsidios para combustibles y modernización de la flota pesquera en méxico. Online, 2018.

[12] Daniel A Revollo-Fernández, Stuart Fulton, and Sara Chávez Sánchez. Value and economic impact of fuel subsidies on the mexican fishing industry. *Appl. Econ.*, pages 1–13, September 2024.

[13] Fernando Aranceta-Garza and Andrés Cisneros-Montemayor. *Learning from the unexpected: informing better policies from a past reform of fisheries subsidies*, pages 93–106. Edward Elgar Publishing.

[14] R Gillett and Food and Agriculture Organization of the United Nations. *Global Study of Shrimp Fisheries*. Food and Agriculture Organization of the United Nations, 2008.

[15] Kristen Dubay, Saori Tokuoka, and Gary Gereffi. A value chain analysis of the sinaloa, mexico shrimp fishery. Technical report, Durham, NC, 2010.

[16] Comisión Nacional de Acuacultura y Pesca. Produjo méxico 47 mil 664 toneladas de camarón en la temporada de captura 2019-2020: Agricultura. `http://www.gob.mx/conapesca/articulos/produjo-mexico-47-mil-664-toneladas-de-camaron-en-la-temporada-de-captura-2019-2020`, 2020. Accessed: 2025-12-30.

[17] Andrés M Cisneros-Montemayor and Tayler M Clarke. Exploración de los posibles impactos de las reglas de la OMC sobre subsidios a la pesca: El caso del camarón en la costa oeste de américa latina. Technical report, International Institute for Sustainable Development, 2019.

[18] Cecilia Sanchez Garcia. Carta de cecilia sanchez garcia al comisionado alejandro armienta mer.

[19] Benjamin S Halpern, Shaun Walbridge, Kimberly A Selkoe, Carrie V Kappel, Fiorenza Micheli, Caterina D'Agrosa, John F Bruno, Kenneth S Casey, Colin Ebert, Helen E Fox, and Others. A global map of human impact on marine ecosystems. 319:948–952.

[20] Fabio Favoretto, Ismael Mascareñas-Osorio, Lorena León-Deniz, Carlos González-Salas, Horacio Pérez-España, Mariana Rivera-Higueras, Miguel-Ángel Ruiz-Zárate, Alejandro Vega-Zepeda, Harold Villegas-Hernández, and Octavio Aburto-Oropeza. Being isolated and protected is better than just being isolated: A case study from the alacranes reef, mexico. *Front. Mar. Sci.*, 7, November 2020.

[21] Sara Avila-Forcada, Adán L Martínez-Cruz, and Carlos Muñoz-Piña. Conservation of vaquita marina in the northern gulf of california. *Mar. Policy*, 36(3):613–622, May 2012.

[22] E A Aragón-Noriega, G Rodríguez-Quiroz, M A Cisneros-Mata, and A Ortega-Rubio. Managing a protected marine area for the conservation of critically endangered vaquita (phocoena sinus norris, 1958) in the upper gulf of california. *International Journal of Sustainable Development & World Ecology*, 17(5):410–416, October 2010.

[23] Koichiro Ito. Do consumers respond to marginal or average price? evidence from nonlinear electricity pricing. *Am. Econ. Rev.*, 104(2):537–563, February 2014.

[24] Miguel Ángel Cisneros-Mata. Some guidelines for a reform in mexican fisheries. *Cienc. Pesq.*, 24(1):77–91, 2016.

[25] A Giron-Nava, A F Johnson, and others. Managing at maximum sustainable yield does not ensure economic well-being for artisanal fishers. *Fish and*, 2019.

[26] A Giron-Nava, V W Y Lam, O Aburto-Oropeza, and others. Sustainable fisheries are essential but not enough to ensure well-being for the world's fishers. *Fish and*, 2021.

[27] Louise S L Teh, Lydia C L Teh, Alfredo Giron-Nava, and U Rashid Sumaila. Poverty line income and fisheries subsidies in developing country fishing communities. *npj Ocean Sustainability*, 3(1):1–9, March 2024.

[28] CONAPESCA. Localización y monitoreo satelital de embarcaciones pesqueras, March 2021.

[29] Juan Carlos Villaseñor-Derbez and Aubriana Longnecker. Hourly vessel tracking data from mexico's vessel monitoring system (vms; 2007 - present): v0.0.1, November 2025. URL `https://doi.org/10.5281/zenodo.17592444`.

[30] Sara Orofino, Gavin McDonald, Juan Mayorga, Christopher Costello, and Darcy Bradley. Opportunities and challenges for improving fisheries management through greater transparency in vessel tracking. *ICES J. Mar. Sci.*, 80(4):675–689, May 2023.

[31] CONAPESCA. Avisos de arribo, cosecha y produccion. `https://conapesca.gob.mx/wb/cona/avisos_arribo_cosecha_produccion`, October 2025. Accessed: –.

[32] Juan Carlos Villaseñor-Derbez and Aubriana Longnecker. High resolution historical (2000-present) fisheries production for small- and large-scale mexican fishers: v0.0.1, November 2025. URL `https://doi.org/10.5281/zenodo.17592475`.

[33] D Lluch-Cota, D Lluch-Belda, S Lluch-Cota, J López-Martínez, M Nevárez-Martínez, G Ponce-Díaz, A Salinas-Zavala, A Vega-Velazquez, Lara Lara, Jr, G Hammann, and J Morales. Las pesquerías y el niño. In Víctor O Magaña-Rueda, editor, *Los Impactos de El Niño en México*, chapter 5, pages 137–178. Universidad Nacional Autónoma de México, Ciudad de México, 1999.

A Supplementary Materials

A.1 Subsidy program description

For the time period analyzed in this study (2011 - 2019), four related fuel subsidy programs in Mexican fisheries have been implemented: *PROCAMPO para vivir mejor* (2011 - 2012), *PROCAMPO Productivo* (2013), *Fomento a la productividad pesquera* (2014 - 2019) and *Subcomponente diesel marino* (2018). The operational rules of the fuel subsidy program in Mexican fisheries are as follows. The fuel subsidy program provides a 2-peso per liter subsidy over a portion of the total fuel used by a vessel, here termed the fuel cap of vessel i ($\hat{Q}_i$). As stated in the program's operational rules[6], the subsidized portion of fuel for any diesel-consuming vessel is calculated using the following formula:

$$\hat{Q}_i = (MDL_i \times DPC_i) \times AF_i \tag{5}$$

Where $\hat{Q}_i$ represents the fuel cap on the subsidy program given to vessel i. MDL_idenotes the"Maximum Daily Liters" of vessel i, and is what the government expects the vessel's fuel consumption to be. DPC_i represents the "Days Per Cycle": the number of days a vessel is allowed to fish during a fishing season. The MDL_i is based on engine size (Figure S1), while DPC_i is determined by the fishery in which the vessel participates[7]. Finally, AF_i is an exogenous adjustment factor set by CONAPESCA and takes values between 0 and 1. This is independent of fishery, engine power, or stock status and is instead determined by budgetary constraints. The adjustment factor was typically set between 0.4 and 0.7, but local officials may downward adjust it. These variations in adjustment factor provide the source of variation that we will use to identify the effect of fuel fishery subsidies on exacerbating overfishing.

[6]See Section 4.1.2 of Acuerdo por el que se dan a conocer las Reglas de Operación de los Programas de la Secretaría de Agricultura, Ganadería, Desarrollo Rural, Pesca y Alimentación, available at `https://dof.gob.mx/nota_detalle.php?codigo=5228713&fecha=30/12/2011#gsc.tab=0`

[7]A fishery is defined as the combination of species and location. For example a vessel targeting tuna in the Pacific ocean is part of the Pacific tuna fishery.

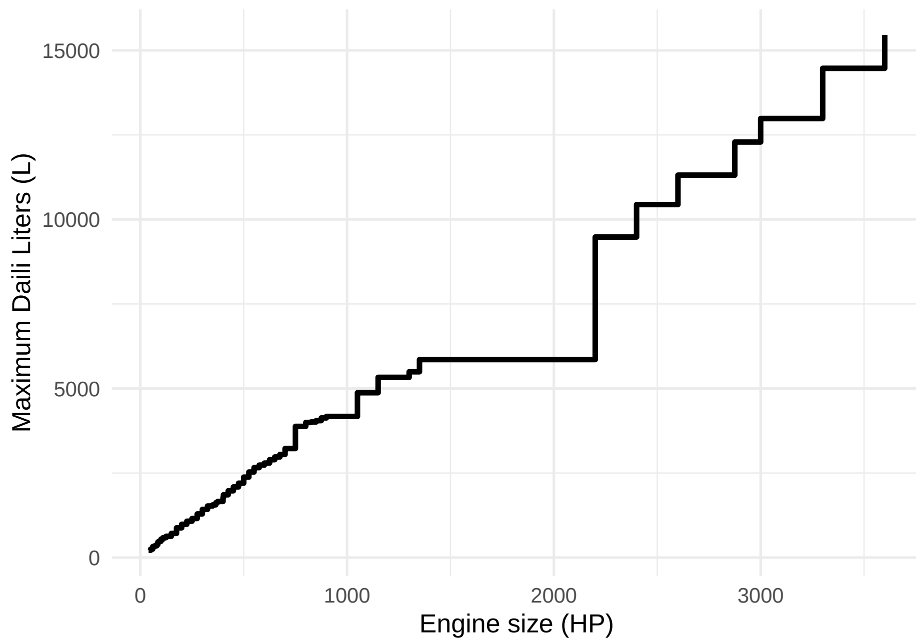

Figure S1: Expected daily fuel consumption for different engine powers. The x-axis shows engine power bins (in HP) as defined by CONAPESCA's operational rules. The y-axis shows the estimated maximum daily liters of fuel to be consumed for the corresponding engine power bin.

A.2 Supplementary Figures and Tables

Table S1: Summary statistics comparing the mean, standard deviation, and range of subsidy amounts and outcome variables across treatment statuses.

	treated	Mean	SD	Min	Max
Subsidy amount (2019 MXN)	Not subsidized	0.00	0.00	0.00	0.00
	Subsidized	790 634.44	1 511 080.57	1521.99	19 633 910.95
Fishing activity (hours)	Not subsidized	1177.34	2405.59	0.00	26 520.33
	Subsidized	4167.75	7053.72	0.00	65 835.78
Fished area (km^2)	Not subsidized	6168.02	9275.14	0.00	92 348.65
	Subsidized	17 966.41	16 374.48	0.00	116 771.44
Landings (Kg)	Not subsidized	9502.86	25 384.33	0.00	285 093.00
	Subsidized	67 204.62	101 070.81	0.00	1 099 556.00

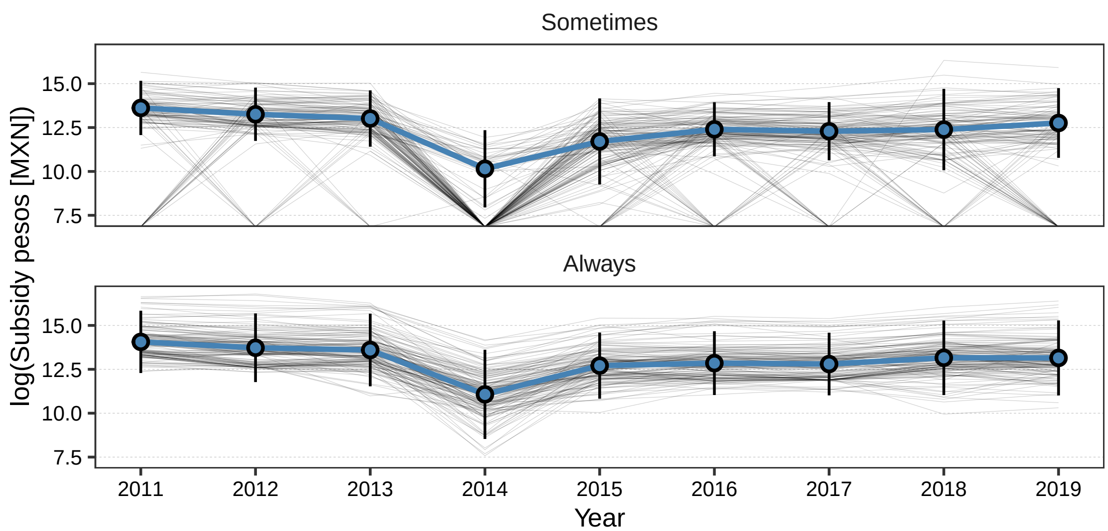

Figure S2: Change in the subsidy amount (Mexican Pesos) granted to each economic unit between 2011 and 2019. The top panel shows data for economic units that are subsidized at least once, the bottom panel shows data for economic units that are always subsidized in our period of study. Each thin black line corresponds to one economic unit. When a line touches the horizontal axis it implies it is not subsidized in that period. The overlaid points show mean ± sd. Note that not a single black line is perfectly horizontal, providing evidence of within-unit variations in subsidy allocations through time.

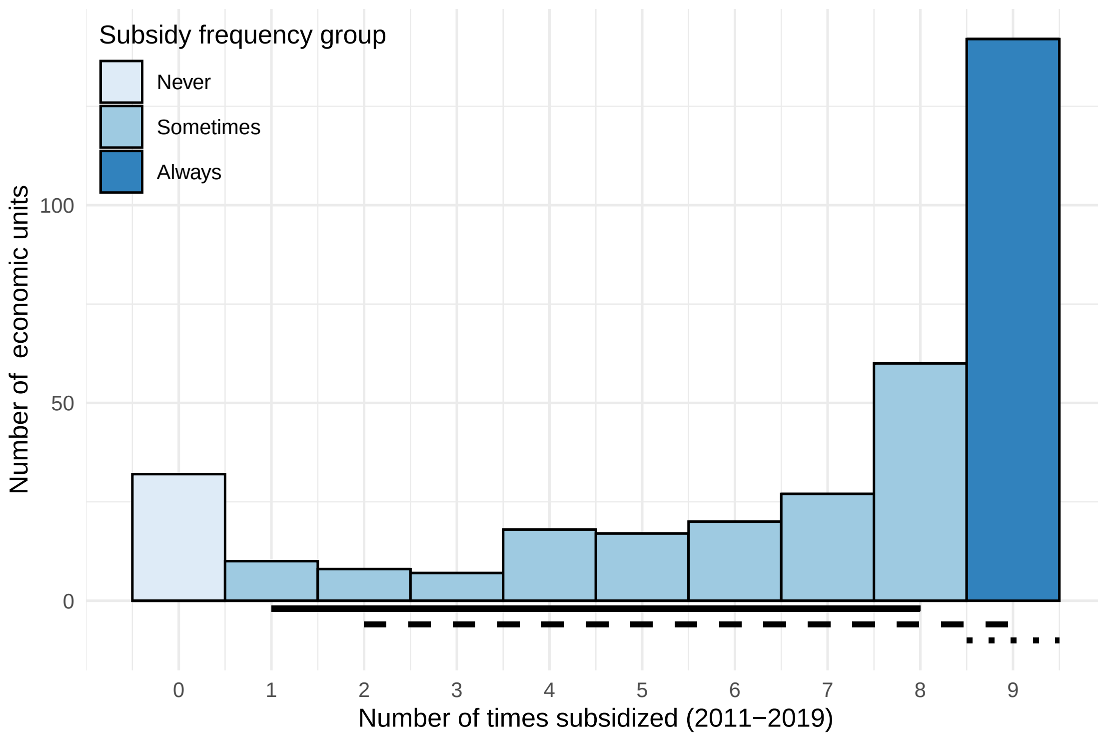

Figure S3: Histogram of frequency with which economic units are subsidized (2011-2019). A value of N = 0 along the horizontal axis implies never subsidized, while N = 9 implies always subsidized. Our main semi-elasticity estimates use vessels sometimes subsidized (i.e. N = 1-8). Our main elasticity estimates use all vessels subsidized N >= 2 times.

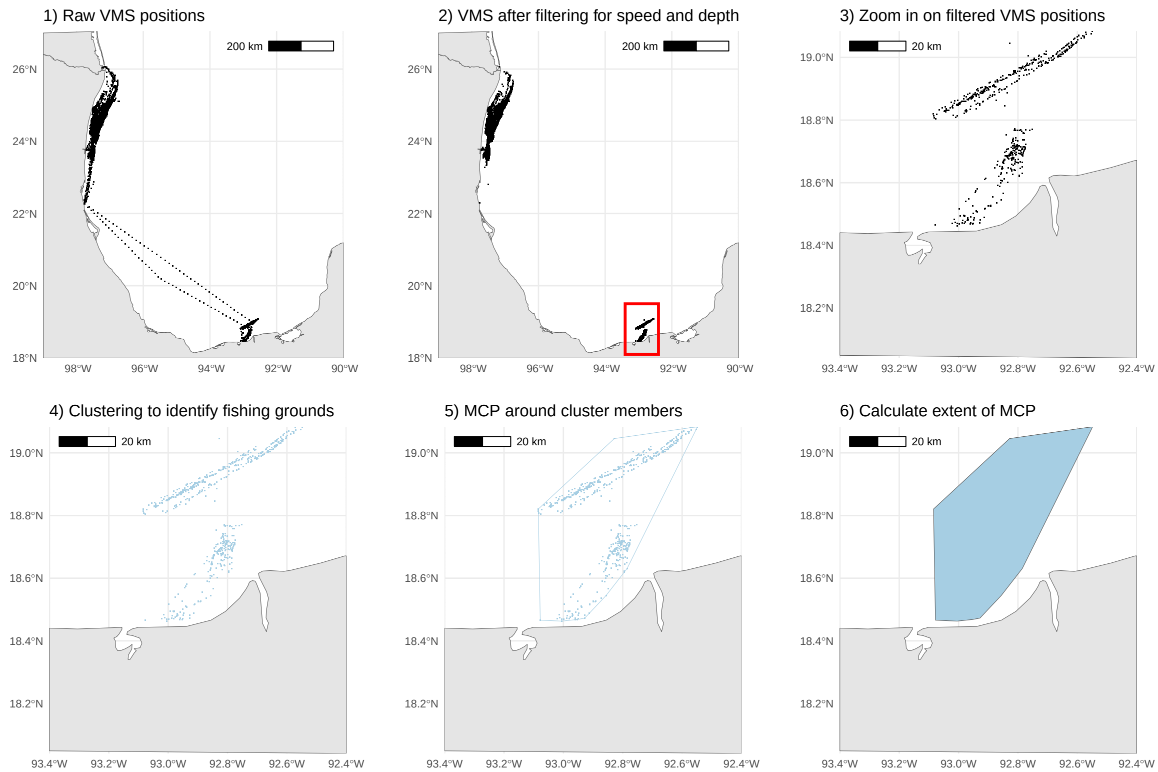

Figure S4: Example showing the process of identifying fishing grounds. Panel 1) shows raw VMS positions for a single vessel. Panel 2) shows VMS positions after filtering for speed and depth parameters. Panel 3) zooms in on the southernmost fishing grounds highlighted in the red square fo panel 2). Panel 4) shows the output from clustering. Panel 5) shows the minimum convex polygon (MCP) around cluster members and Panel 6) builds shows the extent of each fishing ground.

A.3 Robustness tests

A.3.1 Responses to changes in subsidy status

We inspect the robustness of our main-text estimates by performing a series of tests using different specifications and samples. Our first test drops all fixed-effects, and instead incorporates covariates for number of vessels in 2011, total engine power in 2011, log-price of diesel fuel, and nino3.4 index interacted by region. This specification produces in similar coefficient estimates. The second test uses the same specification as our main-text estimates, but instead modifies the sample. Recall that our main-text sample excludes economic units never (N = 32) and always (N = 142) subsidized between 2011 and 2019. This second test includes all economic units, and yields similar coefficient estimates. Our final test removes vessels that were listed in the "fleet modernization" roster, meaning that, in addition to fuel subsidies, they received support to purchase new engines. Removing these vessels yields similar coefficient estimates, leaving our conclusions unchanged. We perform the same battery of tests for our extensive margin estimates (Figure S5 and Table S2), intensive margin estimates (Figure S6 and Table S3), and semi-elasticity estimates (Figure S7 and Table S4).

As an additional check, we test for changes in the estimated coefficients when limiting the sample to vessels subsidized at least 3, 4... 8 times. Figure S3 provides more information on the number of vessels in each sub-sample. We find that the coefficients for time fishing and fished area are largest for vessels subsidized at most 2 times, and gradually converge toward our main-text estimates. The largest coefficient for landings is observed for vessels subsidized just once, but the coefficients also converge toward the main text estimates. Results for extensive margin effects are shown in Figure S8, results for intensive margin effects are shown in Figure S9, and results for semi-elasticity estimates are shown in Figure S10.

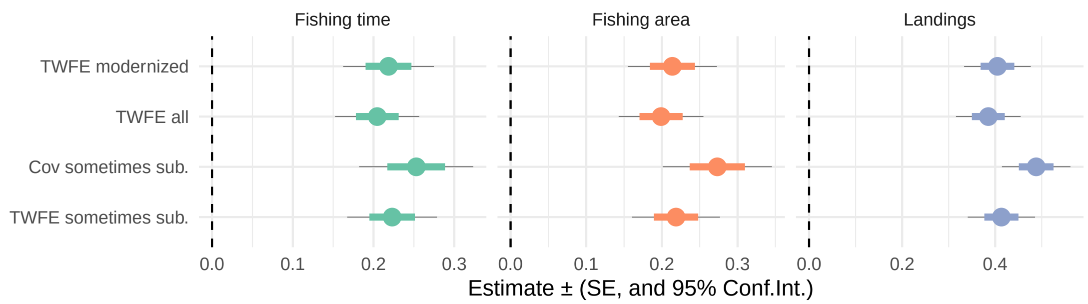

Figure S5: Coefficient estimates for the extensive margin on time fishing, fishing area, and landings with respect to subsidy status for different specifications and samples. Points are coefficient estimates, colored lines show panel-robust standard errors, and black lines show 95% confidence intervals. "TWFE sometimes sub." refers to the main-text estimates, "Cov sometimes sub" refers to estimates for a model specification that swaps fixed-effects for covariates. "TWFE all" refers to the same two-way fixed effects specification as in the main text, but this time including all economic units regardless of their subsidy frequency and status. "TWFE modernized" removes vessels that were recipients of the fleet modernization subsidies. See Table S2 for more details.

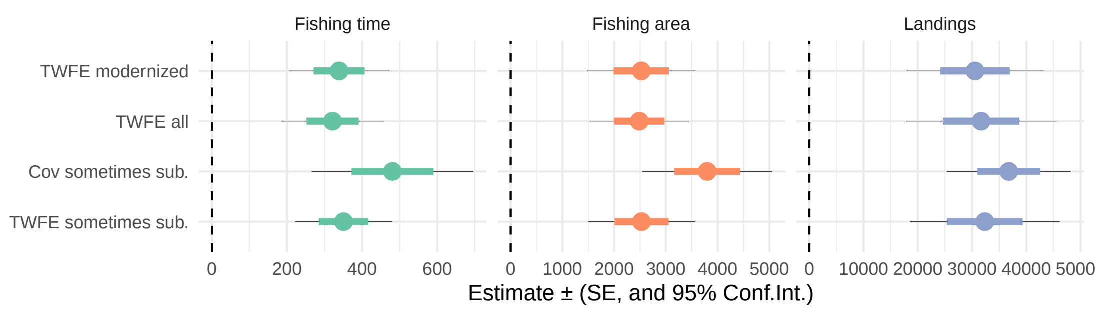

Figure S6: Coefficient estimates for the intensive margin (in levels) of time fishing, fishing area, and landings with respect to subsidy status for different specifications and samples. Points are coefficient estimates, colored lines show panel-robust standard errors, and black lines show 95% confidence intervals. "TWFE sometimes sub." refers to the main-text estimates, "Cov sometimes sub" refers to estimates for a model specification that swaps fixed-effects for covariates. "TWFE all" refers to the same two-way fixed effects specification as in the main text, but this time including all economic units regardless of their subsidy frequency and status. "TWFE modernized" removes vessels that were recipients of the fleet modernization subsidies. See Table S3 for more details.

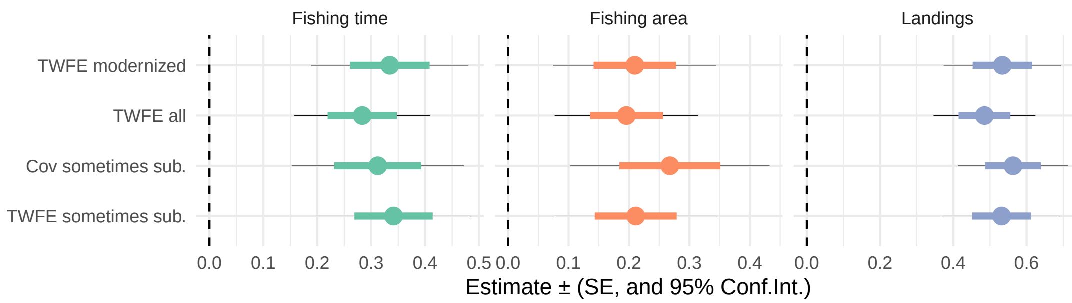

Figure S7: Coefficient estimates for the semi-elasticities of time fishing, fishing area, and landings with respect to subsidy status for different specifications and samples. Points are coefficient estimates, colored lines show panel-robust standard errors, and black lines show 95% confidence intervals. "TWFE sometimes sub." refers to the main-text estimates, "Cov sometimes sub" refers to estimates for a model specification that swaps fixed-effects for covariates. "TWFE all" refers to the same two-way fixed effects specification as in the main text, but this time including all economic units regardless of their subsidy frequency and status. "TWFE modernized" removes vessels that were recipients of the fleet modernization subsidies. See Table S2 for more details.

Table S2: Effect of receiving a fuel subsidy on time fishing (hours) >0, fishing area (km^2) >0, and landings (kg) >0.

	Fishing time	Fishing area	Landings
A) Main text specification			
Subsidized	0.223 (0.028)***	0.219 (0.029)***	0.413 (0.037)***
N	1431	1431	1431
R^2 Adj	0.531	0.547	0.597
B) Covariates but no fixed effects			
Subsidized	0.253 (0.036)***	0.273 (0.037)***	0.488 (0.037)***
N	1431	1431	1431
R^2 Adj	0.206	0.188	0.373
C) Main text specification with all units			
Subsidized	0.205 (0.027)***	0.199 (0.028)***	0.385 (0.035)***
N	2941	2941	2941
R^2 Adj	0.520	0.588	0.703
D) Main text specification without modernized units			
Subsidized	0.219 (0.028)***	0.214 (0.030)***	0.405 (0.036)***
N	1411	1411	1411
R^2 Adj	0.531	0.547	0.603

$*p < 0.1, **p < 0.05, ***p < 0.01$

The unit of observation is an economic unit by year. Numbers in parentheses are cluster-robust standard errors, clustered at the economic-unit level. Panel A) shows the same information as in Table 1A. Panel B) uses the same sample of vessels subsidized at least once, but removes all fixed effects and adds covariates for number of vessels, total engine power, log-price of diesel fuel, and nino3.4 index interacted by region. Panel C) uses the same two-way fixed effects estimation as in A), but includes all vessels in our sample, regardless of number of times subsidized. Panel D) uses the same two-way fixed effects estimation as in A), but removes vessels that received a fleet modernization subsidy.

Table S3: Effect of receiving a fuel subsidy on time fishing (hours), fishing area (km^2), and landings (kg).

	Fishing time	Fishing area	Landings
A) Main text specification			
Subsidized	350.248 (65.644)***	2528.599 (523.001)***	32347.423 (6977.661)***
N	1431	1431	1431
R^2 Adj	0.924	0.792	0.710
B) Covariates but no fixed effects			
Subsidized	480.622 (109.120)***	3796.464 (635.744)***	36753.146 (5792.281)***
N	1431	1431	1431
R^2 Adj	0.752	0.488	0.377
C) Main text specification with all units			
Subsidized	320.923 (69.470)***	2485.116 (487.072)***	31658.543 (7053.213)***
N	2941	2941	2941
R^2 Adj	0.956	0.863	0.915
D) Main text specification without modernized units			
Subsidized	338.614 (68.040)***	2524.252 (532.600)***	30543.733 (6410.022)***
N	1411	1411	1411
R^2 Adj	0.922	0.793	0.712

$*p < 0.1, **p < 0.05, ***p < 0.01$

The unit of observation is an economic unit by year. Numbers in parentheses are cluster-robust standard errors, clustered at the economic-unit level. Panel A) shows the same information as in Table 1B. Panel B) uses the same sample of vessels subsidized at least once, but removes all fixed effects and adds covariates for number of vessels, total engine power, log-price of diesel fuel, and nino3.4 index interacted by region. Panel C) uses the same two-way fixed effects estimation as in A), but includes all vessels in our sample, regardless of number of times subsidized. Panel D) uses the same two-way fixed effects estimation as in A), but removes vessels that received a fleet modernization subsidy.

Table S4: Effect of receiving a fuel subsidy on time fishing (hours), fishing area (km^2), and landings (kg).

	Fishing time	Fishing area	Landings
A) Main text specification			
Subsidized	0.341 (0.073)***	0.211 (0.068)***	0.532 (0.080)***
N	1290	1235	1192
R^2 Adj	0.726	0.700	0.757
B) Covariates but no fixed effects			
Subsidized	0.312 (0.081)***	0.268 (0.084)***	0.563 (0.076)***
N	1292	1240	1196
R^2 Adj	0.305	0.245	0.294
C) Main text specification with all units			
Subsidized	0.283 (0.064)***	0.196 (0.060)***	0.485 (0.071)***
N	2723	2629	2530
R^2 Adj	0.791	0.783	0.846
D) Main text specification without modernized units			
Subsidized	0.334 (0.074)***	0.210 (0.068)***	0.534 (0.081)***
N	1272	1217	1173
R^2 Adj	0.727	0.701	0.756

$*p < 0.1, **p < 0.05, ***p < 0.01$

The unit of observation is an economic unit by year. Numbers in parentheses are cluster-robust standard errors, clustered at the economic-unit level. Panel A) shows the same information as in Table 1C. Panel B) uses the same sample of vessels subsidized at least once, but removes all fixed effects and adds covariates for number of vessels, total engine power, log-price of diesel fuel, and nino3.4 index interacted by region. Panel C) uses the same two-way fixed effects estimation as in A), but includes all vessels in our sample, regardless of number of times subsidized. Panel D) uses the same two-way fixed effects estimation as in A), but removes vessels that received a fleet modernization subsidy.

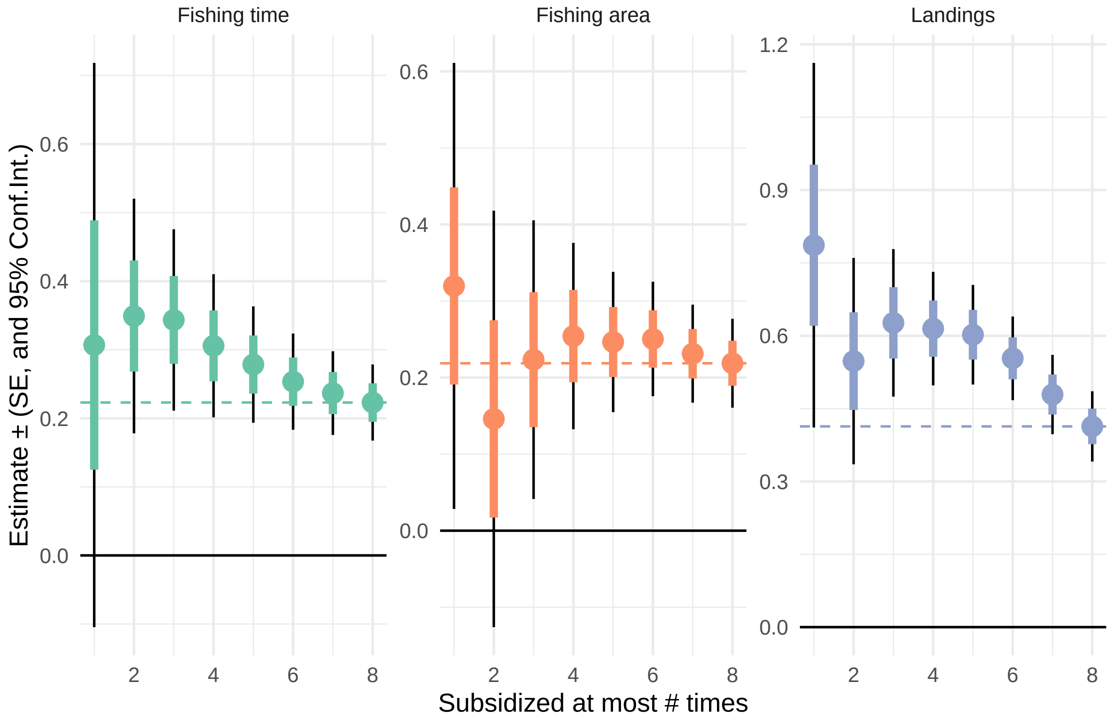

Figure S8: Coefficient estimates for the extensive margin effect on time fishing, fishing area, and landings with respect to subsidy status for different samples based on subsidy frequency. Points are coefficient estimates, colored lines show standard errors, and black lines show 95% confidence intervals. Horizontal dashed lines show the coefficient estimate corresponding to our main specification (Table 1). Each point corresponds to a different sub-sample, where economic units are subsidized at most n times, as indicated by the x-axis. In all cases, the rightmost point (subsidized at most 8 times) corresponds to our main-text estimates.

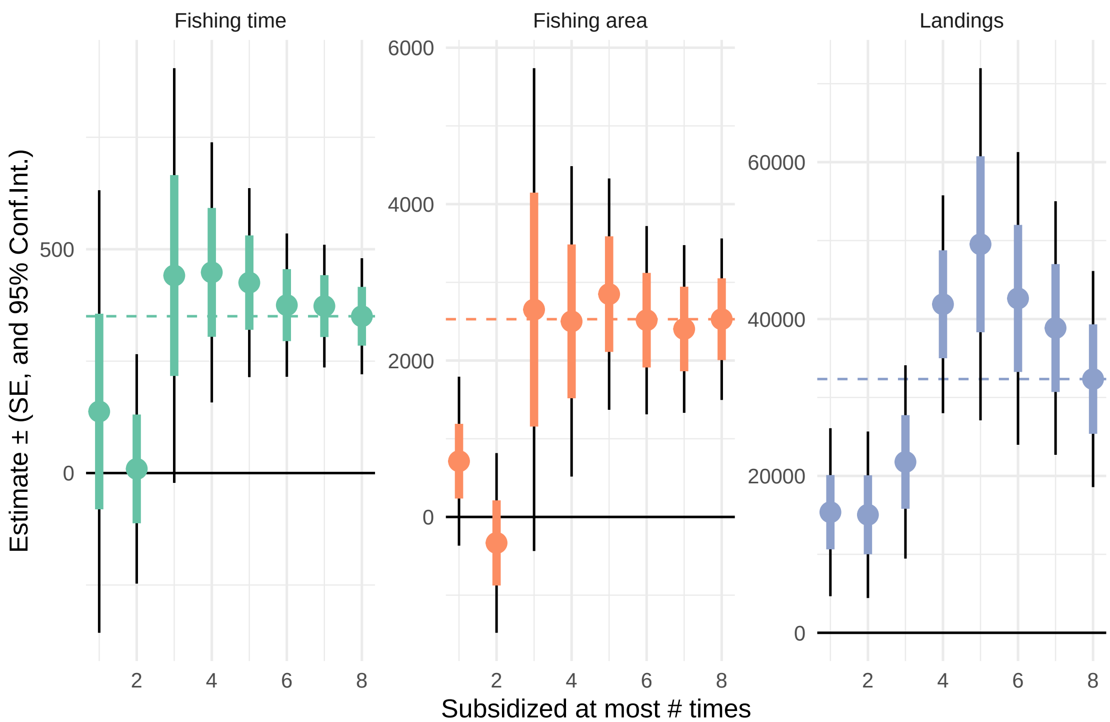

Figure S9: Coefficient estimates for the intensive margin effect (in levels) time fishing, fishing area, and landings with respect to subsidy status for different samples based on subsidy frequency. Points are coefficient estimates, colored lines show standard errors, and black lines show 95% confidence intervals. Horizontal dashed lines show the coefficient estimate corresponding to our main specification (Table 1). Each point corresponds to a different sub-sample, where economic units are subsidized at most n times, as indicated by the x-axis. In all cases, the rightmost point (subsidized at most 8 times) corresponds to our main-text estimates.

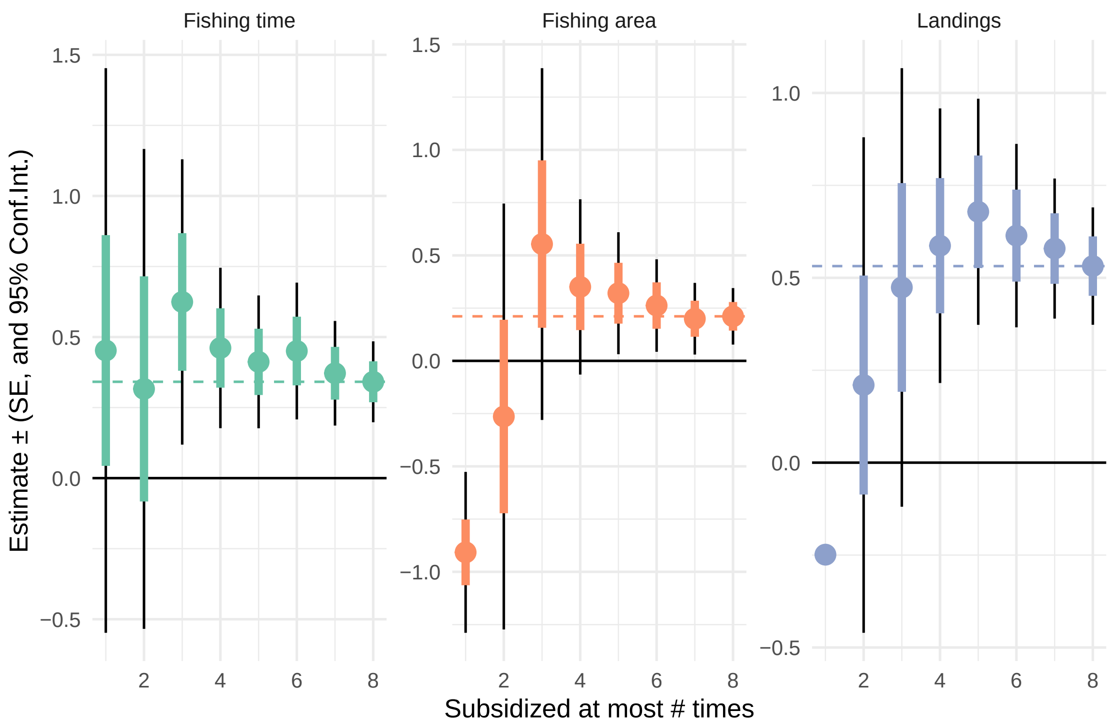

Figure S10: Coefficient estimates for the semi-elasticities of time fishing, fishing area, and landings with respect to subsidy status for different samples based on subsidy frequency. Points are coefficient estimates, colored lines show standard errors, and black lines show 95% confidence intervals. Horizontal dashed lines show the coefficient estimate corresponding to our main specification (Table 1). Each point corresponds to a different sub-sample, where economic units are subsidized at most n times, as indicated by the x-axis. In all cases, the rightmost point (subsidized at most 8 times) corresponds to our main-text estimates.

A.3.2 Responses to changes in subsidy amounts

We inspect the robustness of our main-text elasticity estimates by performing a series of tests using different specifications and samples. The data in Table S5 provides model fit statistics and sample sizes, but coefficients are more easily visualized in Figure S11. Our main-text sample includes all economic units subsidized at least twice between 2011 and 2019 (N = 297). Our first test modifies the sample to include only economic units that were always subsidized during the same period (N = 142). This sub-sample produces similar coefficient estimates, shown in Panel B of Table S5. The next test does the opposite, and excludes economic units that were always subsidized. The results are shown in Panel C of Table S5, and show coefficient estimates similar to those in reported in our main text. As before, the third test removes economic units that participated in the fleet modernization program. This also produces similar coefficient estimates for all outcome variables, as shown in Panel D of Table S5. Finally, we drop our fixed-effects by year and economic unit and instead include the same battery of covariates as before. This specification uses the same sample as our main-text estimates, and results in higher estimates of elasticiy of fishing time, fishing area, and landings at 0.2, 0.19, and 0.37, respectively. These estimates are 40-90% larger than our main-text estimates, and are shown in Panel D of Table S5.

As before, we also conduct one more robustness test where we use our main-text specifications but restrict the sample to economic units subsidized at least 1, 2, 3... 9 times. The results are shown in Figure S12. This test yields stable coefficient estimated across all sub-samples.

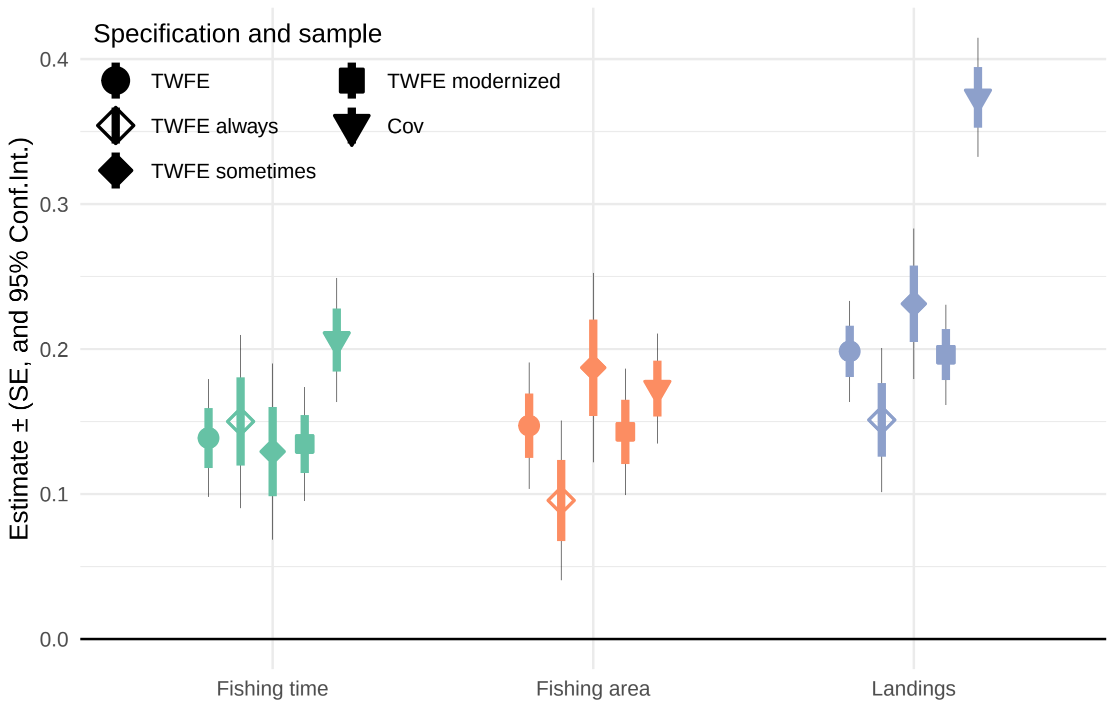

Figure S11: Coefficient estimates for the elasticities of time fishing, fished area, and landings with respect to subsidy amount. Points are coefficient estimates, colored lines show standard errors, and black lines show 95% confidence intervals. The main sample combines all economic units subsidized at least twice between 2011 and 2019. Alternative samples, labeled "Modernized" "Sometimes" and "Always", restrict the sample to economic units that were not part of fleet modernization subsidies, or that are sometimes and always subsidized in the same period, respectively. One-way fixed-effect specifications (labeled "OWFE") drop year-by-region fixed effects and use annual log-transformed mean national fuel prices, the NINO3.4 index values, and a dummy variable for 2014.

Table S5: Elasticity estimates for time fishing (hours), fishing area (km^2), and landings (kg) with respect to changes in subsidy amount.

	Fishing time	Fishing area	Landings
A) Main text specification			
log(subsidy amount[MXP])	0.139 (0.021)***	0.147 (0.022)***	0.198 (0.018)***
N	2240	2202	2246
R^2 Adj	0.850	0.818	0.876
B) Always subsidized			
log(subsidy amount[MXP])	0.150 (0.030)***	0.096 (0.028)***	0.151 (0.025)***
N	1278	1271	1278
R^2 Adj	0.884	0.828	0.905
C) Sometimes subsidized			
log(subsidy amount[MXP])	0.129 (0.031)***	0.187 (0.033)***	0.231 (0.026)***
N	962	931	968
R^2 Adj	0.749	0.746	0.809
D) Removing modernized			
log(subsidy amount[MXP])	0.135 (0.020)***	0.143 (0.022)***	0.196 (0.018)***
N	2208	2170	2214
R^2 Adj	0.850	0.819	0.875
D) Covariates but no fixed effects			
log(subsidy amount[MXP])	0.206 (0.022)***	0.173 (0.019)***	0.374 (0.021)***
N	2242	2206	2247
R^2 Adj	0.517	0.374	0.540

$*p < 0.1, **p < 0.05, ***p < 0.01$

The unit of observation is an economic unit by year. Numbers in parentheses are cluster-robust standard errors, clustered at the economic-unit level. Panel A) shows the same information as in Table 2. Panel B) restricts the sample to economic units always subsidized. Panel C) restricts the sample to economic units sometimes subsidized. Panel D) removes economic units that received fleet modernization subsidies. Panel E) uses the same sample of vessels, but removes all fixed effects and adds covariates for number of vessels, total engine power, and nino3.4 index interacted by region.

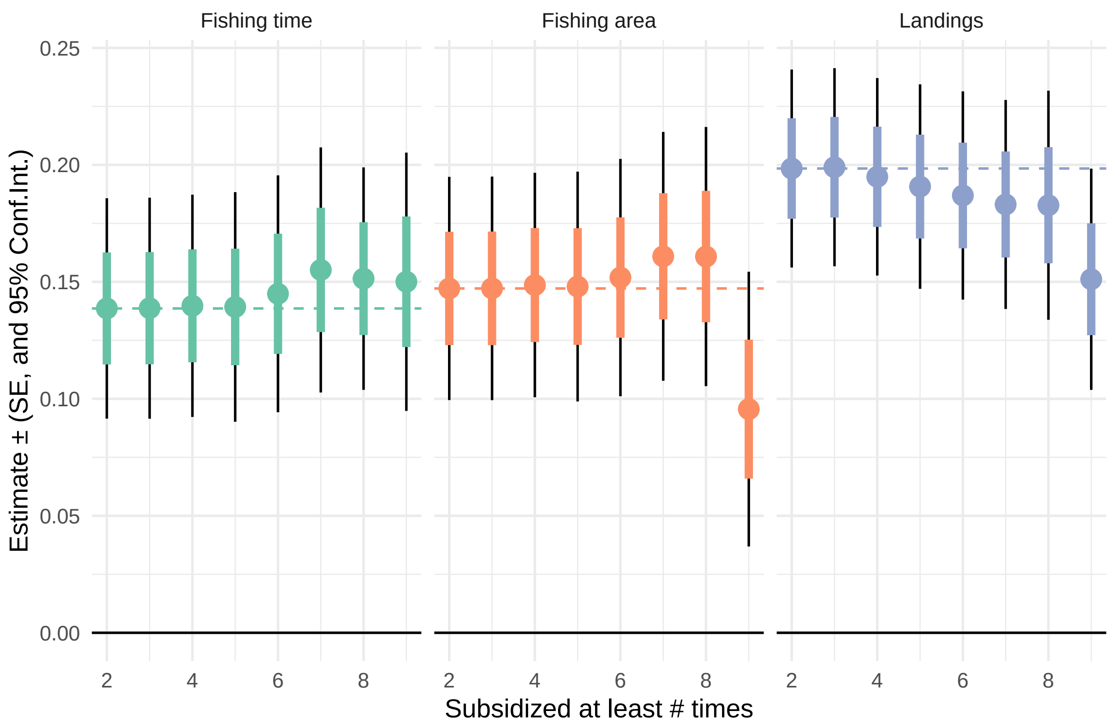

Figure S12: Coefficient estimates for the elasticities of time fishing, fishing area, and landings with respect to subsidy amount for different samples based on subsidy frequency. Points are coefficient estimates, colored lines show standard errors, and black lines show 95% confidence intervals. Horizontal dashed lines show the coefficient estimate corresponding to our main specification Table 2. Each point corresponds to a different sub-sample, where economic units are subsidized at least n times, as indicated by the x-axis. In all cases, the leftmost point (subsidized at least twice) corresponds to our main-text estimates.